MarketScope: Geospatial Analysis For Optimal Small-Business Placement With Integrated Reporting System

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CHAPTER 1
INTRODUCTION

Background of Project

Business location is a decision that entrepreneurs across the globe often have to make, and that is usually done on "feelings" or based on the naked eye alone instead of empirical evidence. This is an area where there is a lack of data-driven decision making which results in a high failure rate of businesses, every single time. Small business owners then face markets blind because they don't have a tool to analyse the gaps and therefore they risk their capital on an already saturated market in which they compete with a wide variety of similar services. To reduce these risks, the use of Geographic Information Systems (GIS) in conjunction with multi-criteria decision making models has become an important strategic tool. For example, you and Zhang used a kernel density estimation and point-of-interest data to find that there are significant differences in the distribution of two competing coffee chains, which can be revealed through spatial analysis in GIS, and which can be utilized to inform the placement of a new coffee chain as a direct result [1]. Similarly, Xu et al. modelled the location decision of retailers under competitive pressures and demonstrated the ability of this type of spatial analysis to differentiate between a location that would take away business from an existing store and one that would serve the actual demand, which is still not met, while not overloading the business [2].

As part of the Filipino archipelagic economy, micro, small and medium enterprises (MSMEs) play a big role in the growth of the country, but they are also very susceptible to operational inefficiencies and poor strategic planning. One challenge is that the data research that is necessary to inform the site selection process for companies is too costly for a small business to execute itself. A case study of a micro-enterprise in Davao City revealed that MSMEs are known to use simple and non-sequential steps instead of a well-structured decision-making process and they are also prone to informal information gathering, which are correlated to the MSME's poor performance and high failure

rate [3]. Strategic planning and market analysis, in addition to being operational functions, also have a direct impact on business success as highlighted in Capiña's assessment of MSEs in Marinduque, which is also constrained by resources for many enterprises despite their potential [4]. Then, the problem of poor placement is more than just an individual business success issue; it must be addressed to help local governments control urban growth and service distribution.

Within the context of Davao del Norte, Panabo is the most important environment for small business placement and competition analysis as it is the main economic hub of Panabo City. A recurring problem in this high activity zone is the saturation of the market, the high-potential zones are not covered and others are very congested. In a study conducted by Alonzo et al. in Misamis Oriental, the most common type of MSMEs had management functions performed by a single person in addition to being a business owner, thus minimizing time and capacity for conducting proper market research or defining a focused target market [5]. This same fragility was evident during the COVID-19 pandemic, as the operations and sales of Davao Region MSMEs were realistically disrupted, highlighting the need to build resilience strategies and the importance of basing these strategies on data rather than assumptions to ensure their survival during times of crisis [6]. MarketScope is designed as an Intelligent Decision Support System (IDSS) powered by geospatial analysis, addressing these local challenges by offering data-driven placements recommendations. MarketScope's inclusion of a spatial evaluation engine and regulatory compliance via City Land Use Plan (CLUP) data provides the average entrepreneur with a level playing field, turning the complex data of geospatial and demographic data into actionable intelligence. Moreover, it employs Generative AI to convert complex spatial variables to accessible feasibility reports.

The study is based in Panabo City, Davao del Norte with context focused on the high activity areas of Panabo Proper. This area was selected because it serves as the business hub of the city, and competition from small businesses is the highest here, where placements are most important. The

project is limited to these boundaries, and most data collection and analysis will be conducted within these boundaries.

In response to these challenges, the project team at the Institute of Computing developed MarketScope, a web-based decision support tool that is optimised for mobile use. It is based on geospatial analysis to produce placement suggestions from real data, not on a guess and check basis, and provides users with a map that provides an overview of the existing distribution of businesses and the risk areas of the environment. This includes a flood zone layer (indicated by red markers on the map) which isn't a mere reference layer but directly influences the calculation of a location's suitability score, meaning the disaster risk component is a key element of the recommendation and not an add-on. The system also inverts the traditional application of a business trend and suggestion: A user is no longer searching for a specific business at a location but the spatial data of an area is read and the categories of MSMEs, that might thrive in this area are suggested. Put another way, MarketScope is taking data that's too complex or too diffuse for a typical entrepreneur to directly use, spatial patterns, infrastructure stand-ins, hazard exposure, and puts it into something that they can go to work with.

This study draws upon a common research and development engagements at the institutional, national and sectoral levels, including the A.C.H.I.E.V.E. Agenda 2025-2030 of the Commission on Higher Education [7]. MarketScope's use of real-time geospatial data to provide business recommendations on mapping, directly operationalizes its Thrust H, Harmonized SDG-Based Research and Innovation, which asks HEIs to undertake SDG-related research and translate research outputs into real-world, usable innovations, and its Enabler I, Integrated Real-Time Data Collection and Analytics System, which envisions using timely and accurate data for planning and decision-making. The study is also guided by a national research and development agenda, the Harmonized National Research and Development Agenda (HNRDA) 2022–2028 [8] which contains a Smart Cities agenda that highlights decision intelligence, prescriptive analytics and Earth Observation solutions for public services, and an Artificial Intelligence agenda that

emphasizes recommender systems and intelligent applications for people to make decisions.

First and foremost, the introduction of MarketScope is well rooted in the institutional development goals of Davao del Norte State College (DNSC) and these are similar to the 2030 Agenda for Sustainable Development of the United Nations. With regard to the institutional level, the study is rooted in the Strategic Priority 3 (Innovation and Inclusion) and the ICT4D thrust (promoting the use of digital tools for grassroots development) of Davao del Norte State College's own Research, Development, and Extension agenda, DNSC SAILS (Sustaining gains in Agri-Aqua to accelerate Innovation and Inclusion Leading to Secured Futures) for 2026-2030 [9], and in the Strategic Priority 2 (Accelerate Agri-Aqua) and Value Chain Optimization component of the same agenda, insofar as MSME site selection supports a broader commercial ecosystem that sustains the local economy of Panabo City. The system makes it easy to access complex geospatial information or market information to aid in the work execution of DNSC's SDG 8 (Decent Work and Economic Growth), and SDG 9 (Industry, Innovation and Infrastructure): foster technological innovation and economic resilience of local MSMEs. In addition, the project reflects DNSC's sustainable thinking and environmental stewardship initiatives, which include SDG 11 (Sustainable Cities and Communities) as a key priority, using City Land Use Plan (CLUP) data and AI-powered scoring to provide actionable insights that drive spatial variables to regulatory compliance, ensuring sustainable and balanced commercial congestion while guaranteeing entrepreneurial investment viability in urban areas. When considered as a whole these agendas form a coherent and supportive chain of alignment that explicitly connects locally relevant contents and outputs of the research to an explicit national (and regional, institutional level) research priorities that DNSC operates for which it is directly responsive to, within the Philippine research and innovation governance structure. The system, in this sense, identifies DNSC's dedication to developing technologies that are directly useful to the local community.

The proponents in Panabo Proper would like to solve a series of related problems that would threaten the survival of local MSMEs. Entrepreneurs often

choose a business location based on intuition or observation and that is evident in the business failure rate. When they can't look at the numbers first they start to crash into each other, and they end up filling up streets that other businesses are filling, and emptying out other streets that are viable. Not only that, but there's a danger side to this that many don't realize—until they lose their inventory and close their business to deal with it until the floodwaters recede—is where they put their business. The proponents are attempting to bridge the gap between geospatial and risk information that exists, and entrepreneurs who don't have access to it or can't understand it, and as a result, open in areas that are both physically dangerous and overcrowded, without realizing it at the time.

Moreover, potential buyers are confronted with a challenge in terms of regulation, disaster risk and planning. Many do not know about the City Land Use Plan or the associated zoning regulations that they are subject to, and may choose a location not even zoned for their business type. Micro entrepreneurs, however, cannot afford to do formal market studies and/or consultancy services. Demographic data at the ground level is also not readily available, and the more practical approach is to use what can be measured: building density, the type of adjoining roads, and the distance of a location from pedestrian-friendly areas where pedestrian activity is naturally occurring, as indicators of demand and site safety. Most of these entrepreneurs do everything from their phone, so that's why the platform needs to be as mobile-friendly as it is desktop-friendly – otherwise, it's not really available for the people who it's supposed to be for.

Objectives of the Study

The primary goal of this study is to develop MarketScope, a decision support system that utilizes geospatial analysis to provide data-driven recommendations for optimal small-business placement in Panabo City.

The specific objectives are as follows:

1. Test and evaluate Fuzzy Logic, GIS-AHP, and TOPSIS algorithms to determine the most accurate and efficient models for a spatial evaluation engine.
2. Design a web-based mapping interface that visualizes existing business establishments, local infrastructure, and an environmental hazard overlay.
3. Develop a spatial evaluation engine tailored for thirteen (13) MSME categories that allows users to assess an enterprise, utilizing building density, road classifications, and proximity to traffic generators.
4. Integrate the Panabo City Land Use Plan into the digital map, automatically cross-referencing user-selected locations against local commercial and residential zoning ordinances.
5. Develop a reporting module powered by generative artificial intelligence that translates the raw map, zoning, and proxy data into clear, context-aware, and downloadable business feasibility reports for prospective entrepreneurs.
6. Develop a proactive business trend and suggestion module that identifies optimal business types for specific areas and user's preferences.

Significance of the Study

The study proposes a new data-driven, geospatial method in small business placement, replacing the "gut-feel" with a strategic and intelligent market analysis. It increases the chances of business success, facilitates balanced urban development, supports the local MSMEs in leveling the playing field, and it is part of the field of localized business intelligence. The following are the study's main beneficiaries:

- **Entrepreneurs/ MSME Owners**. This system will help business entrepreneurs reduce the risks of investing in the wrong business location, identify market opportunities, and generate automated feasibility reports without having to pay for corporate business advisers. This will enhance the confidence in the operation and long term business survival.

- **Local Government Unit.** The system will help the local government to control the balanced development of urban space, avoiding saturation of the market in dense places. It also guarantees that proposed business sites are within the City Land Use Plan (CLUP) and local zoning ordinance.
- **Local Consumers and Residents**. Residents will have a greater supply of local goods and services available to them as businesses will be directed to areas that are underserved and have a high potential. This decreases travel to overcrowded commercial areas and increases overall resource distribution in the community.
- **Future researchers and developers**. The foundation for future research on integrating geospatial mapping and AI-driven decision support systems in local economic development is laid by this study. It can drive innovations that will not only support Panabo City in its bid to enhance its local business environment but also a wider array of municipalities aiming to maximize their local business environments.

Scope and Limitation

The project is limited to the area of Panabo City, Davao del Norte with spatial analysis limited to urban and commercial areas of Panabo Proper. The system aims to support aspiring entrepreneurs and MSME owners with data-based information guidance through the process of store siting to ensure the business is commercially viable and physically resilient to disaster. The prototype's spatial evaluation engine is explicitly limited to evaluating thirteen different types of businesses that are common in Panabo City, namely Coffee Shops/Cafes, Print/Copy Centers, Laundry Shops, Car Washes, Food Kiosks/Stalls, Water Refilling Stations, Bakeries, Small Pharmacies, Barbershops/Salons, Motorcycle Repair Shops, Internet Cafes, Meat Shops, and Hardware/Construction Supplies. The evaluation engine takes into account these sectors according to one enterprise model, which guarantees algorithmic stability and accuracy in location scoring.

The system is designed to use dynamic spatial buffering and incorporate a new Hazard Risk Overlay for informed decision making. This overlay determines vulnerabilities at the location by cross-referencing proposed coordinates with local hazard susceptibility maps. This feature is visually represented on the mapping interface with prominent red indicators, which will actively reduce the suitability scores of locations found in the heavy, moderate, or slight flood zones when evaluating their suitability, thus favoring the selection of locations with disaster resistance. Additionally, the scope encompasses a Business Trend Suggestion page, which turns the conventional search methodology upside down by taking into consideration the existing spatial data and proactively suggesting highly viable business models that are specific to the Panabo City commercial landscape.

The development and implementation of the study, however, have a number of operational and technical constraints. Today's hardware and API limitations prevent the system from integrating live vehicle counts, real-time GPS foot-traffic flow and/or live mobile tracking. Important to note, however, on the environmental risk assessment, MarketScope is not an early warning system as the hazard assessment is based only on static, historical shapefiles and data provided by the local government or disaster management agency and does not monitor live weather events, active flood water levels or biological pest outbreaks. Also, the platform calculates a spatial feasibility score but it is not intended to do an operational budgeting, tax computation or financial forecasting on the accounting level, which would involve return-on-investment or break-even analyses. The accuracy and reliability of the density and hazard maps of the system also depend on how frequently maps are manually updated or the most recently available databases supplied by the local government unit.

Further, the system doesn't gather real-time information about a single consumer's purchases or personal financial records. Instead, it relies on spatial proxies that include building density, road network type, and location of nearby complementary traffic generators and economic anchors, to estimate market demand mathematically. The Automated PDF Reporting module uses third-party Generative AI API (e.g., Google Gemini). Because of this, the language model is

limited to only generating natural-language summaries from the hard spatial, proxy, and historical hazard data provided by the system's backend, and is not designed nor legally empowered to provide financial, real estate, environmental engineering, or legal advice. MarketScope can only be deployed as a web-based application system and is built with the mobile-first responsive design to suite users who access the system through their smartphones. Last, but not least, MarketScope is not a business success tool, it only provides information, comments and suggestions.

Review of Related Literature and Works

This will look at the academic and technological underpinnings required to build a spatial decision support system specific to SMEs. It explains the shift from the traditional business placements based on observation to the modern business placements based on data. This review draws on existing academic approaches towards geographic information systems, proxy-based demand estimation, environmental resilience and automated reporting, and confirms the critical need for an accessible tool to reduce the financial risk of local entrepreneurs. Moreover, it compares the proposed one with other technological applications, to provide a context to the unique potentialities of the proposed system for the local economic area.

Related Literature

Multi-Criteria Decision-Making (MCDM)

Local entrepreneurs have been heavily dependent on observational heuristics in their choice of business locations, which greatly affect the failure rate of the business. Recent studies have shown that there is a definite trend in using GIS in conjunction with multi-criteria decision making models to solve the very complex task of retail site selection. This change is illustrated directly by Ghorui et al. who used fuzzy AHP to arrive at empirically weighted criteria based on the input from decision makers and fuzzy TOPSIS to identify the best of multiple candidate locations, applied to the locations of shopping malls in Kolkata, India, ranking the alternatives [10].

A Fuzzy TOPSIS Approach (Subjective & Linguistic)

Many publications have pointed to the combination of Geographic Information Systems (GIS) and Multi-Criteria Decision-Making (MCDM) paradigms as a major paradigm change from subjective, heuristic, observation-based site selection to empirical, data-driven site selection. The experts' opinion-based Fuzzy Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) and objective, data-based GIS spatial layering methods were evaluated in another key study that compared the best locations for shopping centers in Turkey. The results showed that both models were able to converge to the highly commercial target area (Marmara) but the target areas of the other regions changed because of the different compromises made by the two models, namely, the subjective linguistic bias of Fuzzy TOPSIS and the spatial determinism of GIS. This structural difference indicates the potential for using independent workflows and the critical importance of using an integrated framework that can consider hard spatial data layers and expert socioeconomic weights at the same time. To address this specific literature gap, the architecture of MarketScope directly integrates the capabilities of both data-driven spatial analysis and multi-criteria decision frameworks, forming a unified Intelligent Decision Support System (IDSS) that incorporates multi-criteria geoanalytics, municipal City Land Use Plan (CLUP) zoning compliance, and empirical environmental hazard layers to optimize storefront placement for local MSMEs.

Fuzzy Real Options Valuation (FROV)

Granular street-level census data often does not exist in data-limited settings, and new geospatial frameworks are needed to estimate market demand without relying on a demographic track.In data-limited environments, when granular (street level) census data is not available, new geospatial frameworks are needed that replace the demographic tracking with empirical infrastructure indicators. Problems with local data have been solved with the use of volunteered geographic information (VGI) and open-source spatial data as effective proxies for mapping the distribution of micro-populations and

commercial infrastructure. This relationship is also confirmed by Zhang et al., who use population data as a validation metric for building completeness in OpenStreetMap and show that the distribution of buildings is a good proxy for the spatial distribution of humans, building completeness, and consequently, the spatial distribution of local market demand [13]. Luo et al. apply this concept directly to the distribution of commercial infrastructure, estimating retail density from point-of-interest coordinates and selecting kernel density estimation for mapping the clustering of commercial infrastructure, which they show can reliably separate over- and under-supplied high density commercial zones in a transitioning city further supporting infrastructure density as a viable alternative to market data when it is unavailable [14].

Modern systems can extract these spatial dependence features and provide fine resolution mapping without the need for the traditional, and costly, census survey. In addition to the economic factors, the right selection process for a site should also consider the environmental risks and the bylaws of the municipality. However, Hussain et al. verify that the spatial decision support system is very important in the modelling of natural hazards, using GIS-based multi-criteria analysis along with machine learning to evaluate the flood resilience and they show that there is a significant increase in long-term disaster preparedness when using the evaluation of proximity to flood-susceptible areas [15].

Related Works

Web-Based Point of Sale (POS) Systems

Modern retail stores use Web Based Point of Sale (POS) to protect every day's transaction details, and that is the most frequent operational transaction architecture that's found within these stores. Edison et al. refer to these applications as mainly local financial digitization solutions whose purpose is to enhance the checkout procedure and to keep inventory information up to date for small businesses [16]. This means that although the development of cloud-based POS systems is more focused today, it is largely on backend performance and cost-efficiency, response latency, throughput and infrastructure cost rather than

what a business might require prior to opening its doors [17. This is not a surprising trend: Tudor et al. [18] reviewed almost 5000 publications related to retail marketing to determine the extent to which GIS based techniques are present, observing that GIS based approaches are still not widely used throughout the more common systems and literature used by retailers on a day-to-day basis. That means that standard POS structures are deployed once operational and only gather business performance metrics after the storefront is open; that means backend data repositories, no matter how impressive, have never been designed with geographical considerations in mind, let alone spatial parameters, and never designed to evaluate competitor density prior to the deployment of storefronts that MarketScope is intended to remedy.

The Ideafy AI Platform and Cognitive Brainstorming Tools

Artificial Intelligence in entrepreneurship education and start up incubation has also encouraged the conceptualisation of platforms such as Ideafy AI, an AI tool capable of analysing business concepts and creating feasibility reports for fledgling entrepreneurs [19]. A similar trend was observed with generative AI being adopted as a "co-pilot" for business decision-making processes by helping individuals organise their ideas and test and challenge their current assumptions at the concept stage of a project, according to Bass et al. [20]. Aside from that, however, they all work with text – they take in text prompts and return text as output – and they were never designed to read or reason about spatial data. That's the thing with platforms like Ideafy: They deal with the business data entirely as unstructured text input, and have no way to take into account the ground-up, local variables, building footprint densities, flood hazard zones or proximity to infrastructure that can legitimately decide whether or not a particular storefront is actually going to work in or near a particular locale.

Local Government Unit (LGU) Permit Trackers

E-governance solutions like Local Government Unit (LGU) permit tracking and the e-permit application are intended to address document processing within the context of New Business Registration in the municipality. One such system

was designed by Oñate et al. for two municipalities in Camarines Sur, where the key component of the system was to convey an application for a licence to the required regulatory department processes and approvals that a business application would traditionally need to go through in person [21]. In a similar system, but for other projects, Amer et al. identified that developing these platforms is having workflow routing or shuttling an application from one stakeholder to another until a permit is issued [22]. What both the studies have made clear is that these systems, despite their ability to interface with the municipal master plans or use the zoning boundaries to determine whether the location legally supports a business, are still limited to a regulatory dimension: They assess if the business can legally operate at a given location but were not intended to assess whether the location is financially feasible to begin with. They do not have an analytical engine on board that could decompose road types, assess building intensity, and calculate a suitability measure – they would need to find the same physical gaps without the benefit of any data-based finding mechanism provided by the system which they were supposed to formalize their business with.

Table 1. Comparative Table

Features	Web-Based POS	Ideafy AI Platform	LGU Permit Tracker	MarketScope
Geospatial Visualization			✔	✔
Automated PDF Reporting	✔	✔	✔	✔
Decision Support Score	✔	✔		✔

Zoning Integration			✔	✔
"Market Gap" Identification		✔		✔

From the table 1, MarketScope is the only system that includes geospatial visualization, zoning integration and market gap identification. Web based POS, Ideafy AI and LGU permit trackers offer automated reporting or decision-support capabilities for general business management or administrative processes but are not developed to collect spatial proximity data to use for pre-deployment site selection.

Web-Based POS

Web based Point of Sale (POS) systems are the typical transaction architecture that is widely adopted by retail stores to handle transactions and inventory management. Though Web-Based POS systems provide automated PDF reporting and digital financial records for operations, they are based on a general transaction model and are not a pre-deployment spatial analysis tool. According to described by Edison et al., Web-based POS systems are used more as a financial digitalization solution where they do not gather or showcase geographic market information used for site selection [16]. Web-Based POS systems also lack the local market gap and competitor proximity calculations and display, which is a key analytical component of MarketScope for local entrepreneurs.

Ideafy AI Platform

Ideafy AI Platform is an artificial intelligence application that helps entrepreneurs analyze and conceptualize a startup business. The Ideafy platform has business ideation capabilities and generates a decision support score as a function of text input, but according to Shah et al.'s report on the platform, it is

primarily focused on natural language generation and cognitive brainstorming without any localized spatial zoning integration or any physical proximity metrics [23]. The Ideafy AI Platform is a generic text generation platform which isn't a specific tool for identifying and designing environmental hazard intersections, zoning compliance or even physical market gaps within a defined urban ecosystem like Panabo City.

LGU Permit Tracker

The localized e-governance platforms are being developed by the municipal governments to ease the business registration process, called LGU Permit Trackers. These platforms are based on a workflow routing model, which enables the local government units to handle their documents, and track application statuses in real-time. This model is good for integration and administration, but it is not competitive in terms of spatial analysis or market research. Both studies, by Oñate and Amer [21] confirm that e-permit systems are structured around the validation process of an application, with no exception for infrastructure to gather and present geospatial visualization, competitor density or multi-criteria decision support scores calibrated for business viability. MarketScope will not solve the pre-deployment operational data-gaps that LGU Permit Trackers focus on for local SMEs.

Definition of Terms

Artificial Intelligence (AI) - Artificial Intelligence is the ability of computers or computer-based systems to mimic human intelligence processes, including machine learning, natural language processing and generative capabilities, to undertake complex cognitive tasks.

Building Density - A measure of building concentration in urban planning, usually expressed as building surface area or building roof area per area of land zone.

Business Trend Suggestions - A forward-looking spatial analytics function that analyses unserved market voids in a given region, consumer signals, and geographic trends to provide highly viable business opportunities dynamically for a targeted location.

Catchment Radius - The geographic area or service area around a commercial point of interest where a business naturally draws its business and has competition issues immediately.

City Land Use Plan (CLUP) - An overall policy document and regulatory framework established by the municipality that outlines the official long-term zoning classifications of commercial, industrial, residential and agricultural land uses.

Flood Zone Layer - A locally developed environmental mapping dataset, showing the geographic extent to which environmental hazards resulting from flood events are of varying impacts dependent on environmental evaluation criteria.

Geomarketing - Specialized subset of marketing and spatial economics that combines geographic intelligence, location-based data, and mapping technologies to provide the commercial advantage, capture consumer demand and inform sales territory planning.

Geospatial Analysis - The holistic approach of collecting, transforming, visualizing and analyzing geographic data and spatial patterns to gain understanding of complex physical relationships within a given geographic area.

Hazard Risk Overlay - Hazard susceptibility mapping in a conceptual and digital format overlaid onto base geographical layouts to assess, cross reference and identify physical vulnerabilities and structural exposures to natural hazards.

Hybrid Blending Logic - An algorithmic computing routine for calculating, blending and outputting the combined mathematical feasibility and risk profile of **two separate, co-located variables.**

Intelligent Decision Support System (IDSS) - Highly interactive, intelligent software architecture that combines computational knowledge-bases, expert data-models and artificial intelligence to support human decision makers in poorly-structured, complex problems.

Micro, Small and Medium Enterprises (MSMEs) - Any business or commercial organization primarily based on the total asset size and scale of capitalization, which are important sources of local macroeconomic employment.

Multi-Criteria Decision Analysis (MCDA) - A structured mathematical branch of Operations Research that is used to evaluate, rank and prioritize multiple conflicting criteria using objective weight values for a complex decision problem.

OpenStreetMap (OSM) - A global GIS initiative that is being developed by a community of volunteers, the goal of which is to create a free, editable, openly licensed GeoData set for the planet.

Spatial Proxies - Other observable geographic data points that are used as mathematical equivalents to estimate unrecorded or private socio-economic data, such as the corridor density of a region to estimate its purchasing power.

Suitability Score - a quantitative model-based numerical calculation to rank the developmental/commercial viability of a targeted site based on weighted criteria.

Web-Based Mobile-Responsive Design - An adaptive software interface design technique that allows for a single application developed using web-browser architectures to scale, reposition and optimize user interface breakpoints throughout a wide range of desktop and mobile hardware screens.

CHAPTER 2
METHODOLOGY

In this chapter, the systematic process used in the development of the MarketScope: Geospatial Analysis for Optimal Small-Business Placement with Integrated Reporting System is presented. It outlines the procedure, process and techniques employed by the proponents to develop the system in an organized, efficient and reliable way. The methodology is basically used as a guideline in realizing the study objectives, especially in developing a data driven decision support system to help the entrepreneurs to determine the optimum location of their businesses.

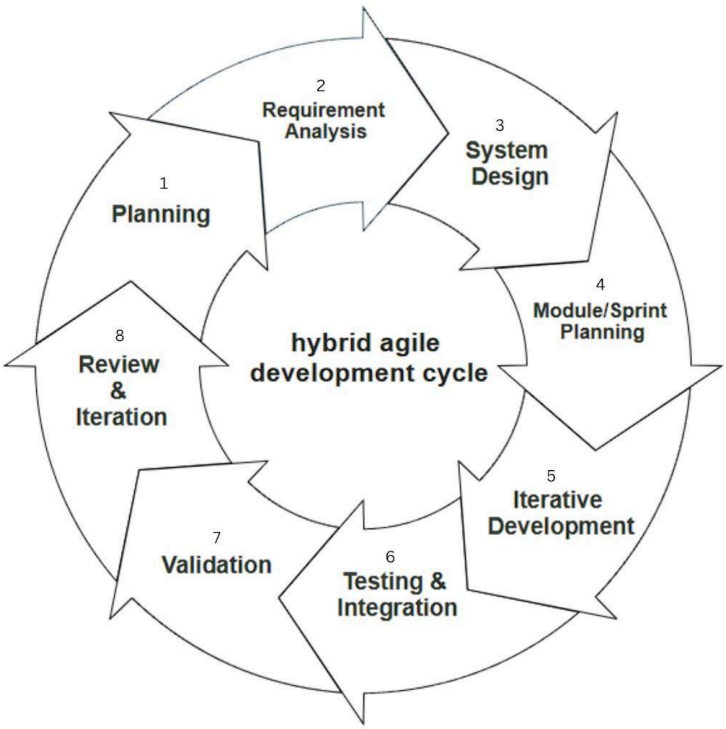


Figure 1. Process of Modified Hybrid Agile Development Model

System Planning

The issue identified during the planning stage was that the lack of data oriented methods on poor location decision of MSMEs in Panabo City.

They started formulating their project by setting its boundaries, scope, users of the project which were business owners and entrepreneurs in the area of Panabo Proper. The team identified their goals, including: building a visual map of business success areas; discovering opportunities in the market; hazard risk assessment; and feasibility reports using artificial intelligence.

The decision to use a hybrid agile approach for the development of MarketScope was made because this project needed both the structure of traditional software development, and the flexibility of iterative, feedback-driven cycles. Structured and upfront phases, Planning, Requirement Analysis, and System Design, were followed by iterative Module/Sprint Planning and Development, which is in line with the practice of hybrid agile that combines a Waterfall approach with sprints as described by Yahya and Maidin as the most important and defining character of hybrid agile practice: software teams that combine a Waterfall-style foundation with Scrum-style sprints to balance predictability and adaptability midway through a project [24]. The system, spanning a series of interdependent components, geospatial analysis, zoning compliance, hazard risk overlays, and AI-driven reporting, found its architectural bedrock in this structure, particularly suited for MarketScope. Meanwhile, empirical evidence backs hybrid methods over traditional ones: Gemino et al.'s international research on 477 cross-industry projects showed that greater satisfaction with the project was achieved when using hybrid methods over traditional ones, while keeping budget, scope and quality steady; more than half of the projects they studied already used some form of hybrid project [25]. Leong et al. support this at the software level by proposing that a software development approach that combines agile sprints and activities with a traditional software development lifecycle enables a team to maintain clear milestones and documentation while remaining responsive to the type of evolving requirements, data sources, newly refined scoring criteria, adviser feedback, that a single developer capstone project may have experienced throughout its development lifecycle [26]. The hybrid model allowed the team to nail a core architecture early in the game without limiting each module, mapping, scoring, zoning, hazard analysis, and reporting to being a discrete sprint to test and refine.

Requirements Analysis

Requirements were gathered from the functional and non-functional perspectives, based on the identified requirements, during the Requirements Analysis stage. The functional requirements included features like visualizing business location in a map, measuring the level of business saturation and demand, adding information about zoning from City Land Use Plan (CLUP), creating hazard-prone areas, such as the areas prone to flooding, creating suitability ratings, and producing feasibility report generated by AI. The non-functional requirements, meanwhile, referred to ensuring that the platform was mobile-friendly, user-friendly, fast and effective data processor. The proponents also identified the critical data sources including OSM, CLUP zoning info and hazard maps.

System Design

The system architecture has been designed and planned for the system to be able to include different parts, which together would provide the desired functions of the MarketScope project. The front end will be an interactive map and user interface while the back end will deal with computation and processing. The system makes use of a database to handle the spatial data, businesses, and zoning, and an artificial intelligence system to create feasibility reports. Using the modular design approach, the following modules are designed to be created: Geospatial Visualization, Spatial Evaluation, Zoning, Hazard Risk Overlay and Artificial Intelligence Reporting.

Module / Sprint Planning

The system was broken into smaller more manageable pieces using an Agile sprint approach. Every sprint emphasized on particular functionalities within the system to enable effective and well-organized development. Sprint 1 focused on the map interface and geographic visualization components and Sprint 2 focused on the business density and business competitor map components. In

Sprint 3, suitability score analysis was addressed, based on multi-criteria decision analysis engine and in Sprint 4, the zoning functionality of CLUP was addressed. In Sprint 5, the focus was on hazard risk analysis, particularly flood zones, and report creation with AI technology was covered in Sprint 6.

Iterative Developement

To develop the system, an iterative development was used, in which the features were improved at the end of each iteration (sprint). The iterative process involved programming of system capabilities, integration of data layering (space, scoring algorithm) and, enhancement of user interface design. Iterations made it possible for the developers to test and assess system performance. Feedback collected in the initial testing stages was used to enhance the accuracy, efficiency and functionality of the system.

Testing and Integration

Each module was thoroughly tested before being installed in the system to verify the compatibility and proper operation of the system. A number of tests were performed, such as unit testing to check the separate components (scoring engine), integration testing to make sure that various modules work together well, system testing to test the overall performance of the system, and user testing to test the usability and performance in terms of the user. All the modules have been successfully tested and integrated into a simplified system to allow the system to operate without problems and to yield stable results.

Validation

To guarantee the quality and relevancy of the results, the system was validated to ensure that it is in line with the aims of this research and produces credible results. System validation involved comparing the system output with expected output; analysing the accuracy of the suitability ratings given by the system; analysing output generated for zoning compliances and analysing the relevance of reports generated by AI technology.

Review & Iteration

The system was continually reviewed and refined following its validation stage in order to improve its efficiency and effectiveness even further. This included the identification of various shortcomings and problems in the existing system, improving the accuracy of the algorithms used in the system, improving usability of the user interface, and overall enhancing the performance of the system.

Project Team Organization

The roles and responsibilities will be explicitly outlined in the project development of the system and will be critical to the successful implementation of the project. A systematical process will be conducted, where the work will be distributed rightly on the team and will be executed efficiently.

The organizational chart of MarketScope shall be displayed in Figure 2:
The aim of the study is to develop a geospatial analysis for optimal placement of small businesses with integrated reporting system. The overall responsibility for the completion of the whole development process belongs to the project manager who will coordinate all the activities of the development process, following the development goals and end-user requirements, respecting the system's design and development.

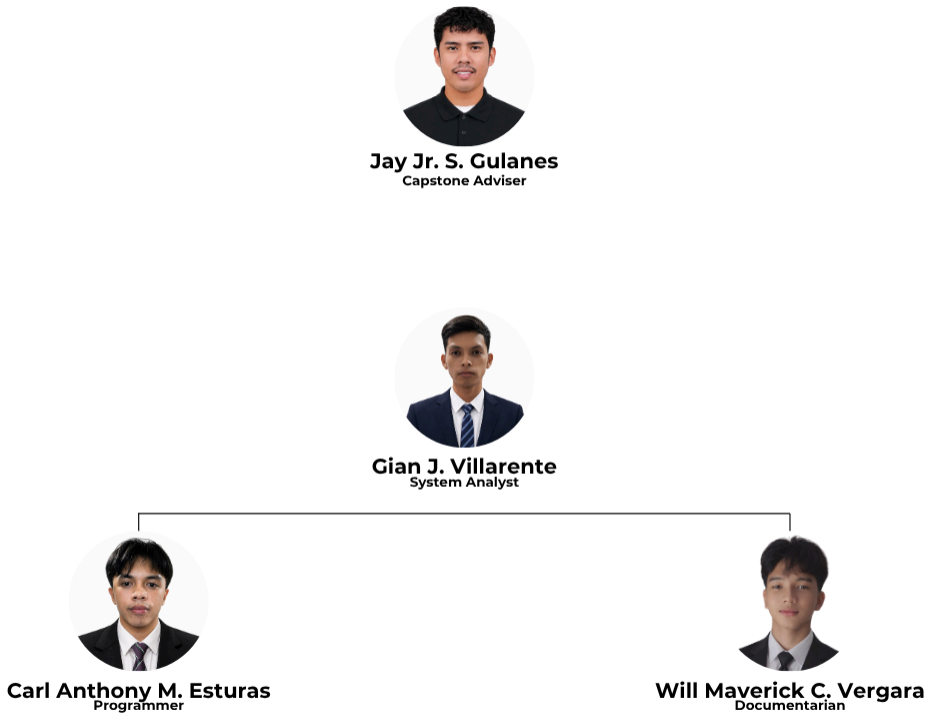


Figure 2. The Project Development Team Chart

The task of the system developer is to design, develop and implement the system front end as well as back end. He/She will implement the geographical information systems and zone information and hazard layers into the software. Moreover, he/she will need to create the spatial assessment engine as well as the artificial intelligence reporting module. The developer shall ensure that all functionalities of the system such as mapping, analysis, scoring and reporting interact with each other successfully.

The documentarian will take charge of collecting, organizing and keeping all project related documentation. The documentarian will prepare all necessary technical documents, system documentation and user manuals.

Work Breakdown Structure

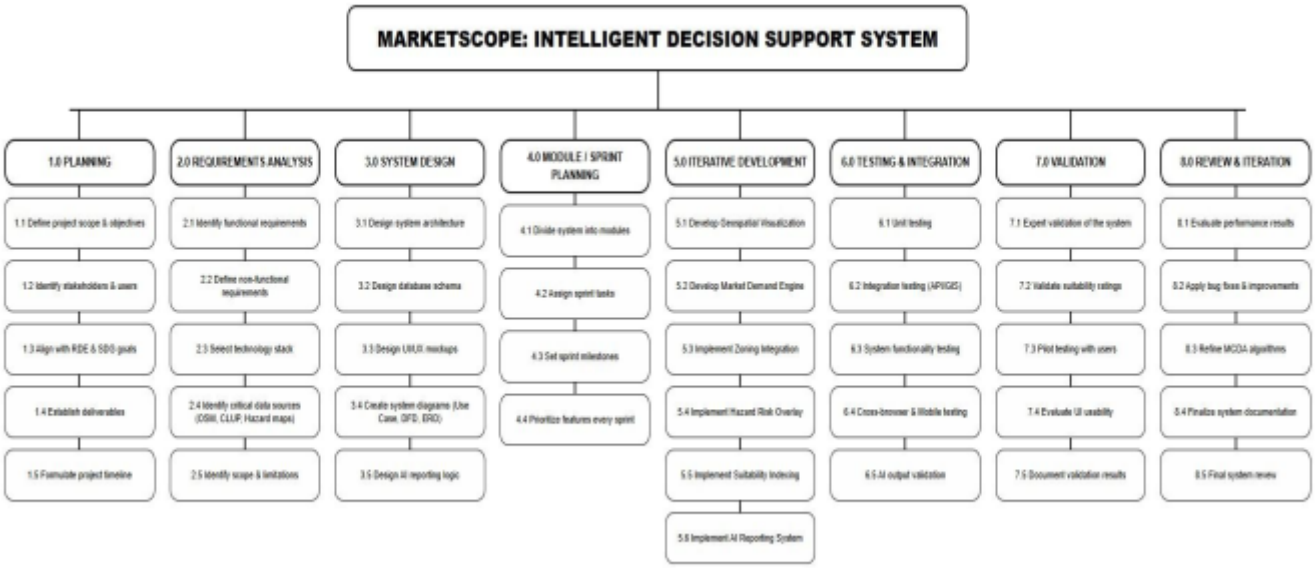


Figure 3. Work Breakdown Structure of MarketScope

In Figure 3, the MarketScope project is presented in the hierarchical and structured Work Breakdown Structure (WBS). All the sub-processes and phases will be broken into the entire project to develop and manage all the elements of the system efficiently. The WBS tool will help structure the project development, making it visible and efficient, planning and time management throughout each sprint phase. The key stages of the project will involve a number of tasks, including the following: Geospatial Mapping,

Using Artificial Intelligence (AI), Business Density Analysis, Zoning Integration, Hazard Risk Assessment, Suitability Scoring and Report Generation. Using the WBS approach in the development of the MarketScope application would enable the implementation of the Agile Iterative Development model with a logical and efficient approach in performing all development processes.

Gantt Chart

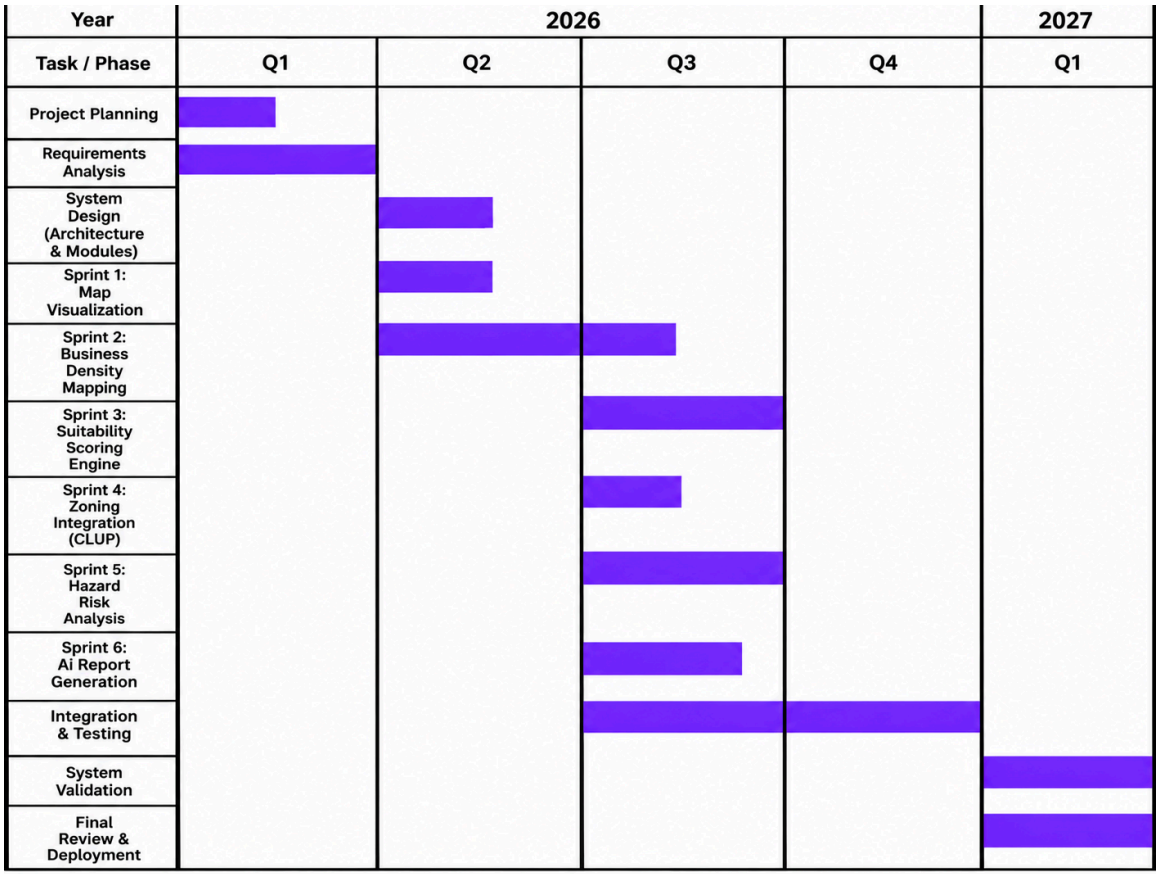


Figure 4. Gantt Chart of the Project of MarketScope

Figure 4 provides a sample of a structured, hybrid project timeline for a specialized software development project over four quarters, for a project that may be an urban planning tool or a geographic information system (GIS). The work starts in Q1 with Project Planning and Requirements Analysis, making rapid progress through a long and protracted System Design stage, interweaving with the first few increments of Development. Development works in stages, as you can see in this image (Figure 4), and starts with core functions (towards the end of the first quarter) all the way up to analytical stages such as Hazard Risk Analysis and AI Report Generation (mid-way through the third quarter). The amount of parallel scheduling is absolutely large; especially during the second and third quarter, when several development sprints are executed in parallel and

the third quarter is dedicated to an extensive Integration & Testing phase. The project culminates in the last part of the third quarter with a special period for System Validation, and the rest of the fourth quarter is left for Final review & Deployment (Final QA right before launch).

System Analysis

System analysis involves defining the problem(s) and stating the requirements for the development of the MarketScope system. The findings from Chapter 1 indicate that when it comes to business location, most MSMEs and entrepreneurs rely on their intuition ("gut feel"), which results in poor business location decisions, overcrowding in the market and increased chances of failure. Hence, it is required to have a system which will utilize the data to make better decisions for the location of businesses. This is where systematic analysis of the proposed solution is important, ensuring it resolves all these problems.

From our study of the current scenario, it is quite evident that there is no integrated solution to address the geospatial and zoning data as well as hazardous area issues within Panabo City. There are no systems available for current business entrepreneurs to use to assess the level of business rivalry in markets, and even hazards in the environment such as floods and earthquakes. Therefore, there is an urgent requirement to develop an intelligent system that is able to integrate complicated datasets to useful decision support mechanisms.

The proposed market scope system will address these gaps by uniting geospatial analysis, decision-making using multiple criteria and the application of AI reporting within one application. The system considers several factors including business density of the area, accessibility of the infrastructure, zoning and hazards, to calculate the suitability score of the selected site. Furthermore, MarketScope enables users to analyze the results on the maps, and also automatically generates a feasibility report. Using it, the spatial data can be transformed into meaningful information in order to make strategic decisions for the location of businesses.

Overall, this system analysis shows that it is necessary to develop an intelligent system to ease decision making for MSMEs. The analysis has identified the challenges with the current methods and the advantages of incorporating technology into business planning.

System Architecture

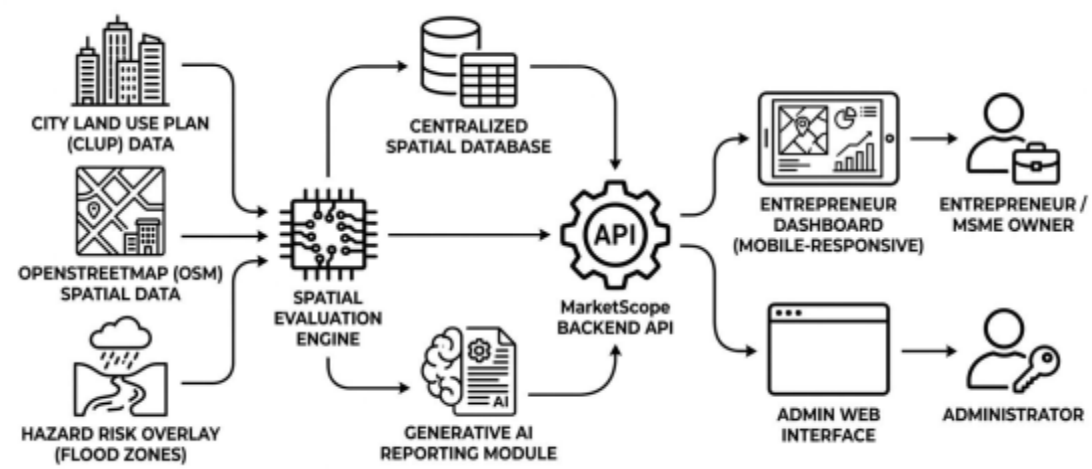


Figure 5. System Architecture of MarketScope

System architecture consists of a number of interrelated layers, which are needed to perform geospatial analysis, data processing, and automatic reporting. Data Layer includes data that is gathered by the system including existing business locations, road networks, hazard risk maps, infrastructure data, CLUP zoning. Processing Layer is tasked to analyse the information collected through geospatial analysis, market demand and competition, perform a zoning analysis, assess hazard risks and determine a suitability score. The information is then handled by Backend Layer in conjunction with the connecting of the database, spatial evaluation engine and AI reporting module. Application Layer includes the web-based, responsive interface that includes a business location selection, a business category selection, map results, zoning/hazard evaluation and

feasibility report generation. Access to the application is controlled by using user authentication and the administrator can manage data, system records, reports, while regular users can analyse locations and export results to the desired formats.

Conceptual Framework

Figure 6 presents the basic operational process of the MarketScope system, depicting the orderly process of how raw administrative and user generated information are converted into comprehensive decision-support intelligence. The input layer consolidates critical information on entrepreneurial site needs and preferences, along with administrative regulatory policies such as CLUP zoning and custom business parameters to provide a foundation for the analysis that is both commercially relevant and in line with municipal regulations. The central processing unit, with two main functions: Geospatial analysis and Business Trend analysis, is the logic engine that brings together these various variables and generates one comprehensive assessment. Finally, the output layer provides the system with its value proposition by presenting a detailed Viability Report and an interactive Mapping Interface that transform complex algorithmic outputs into an intuitive, actionable and visually accessible format, effectively closing the gap between raw data and informed MSME site-selection decisions.

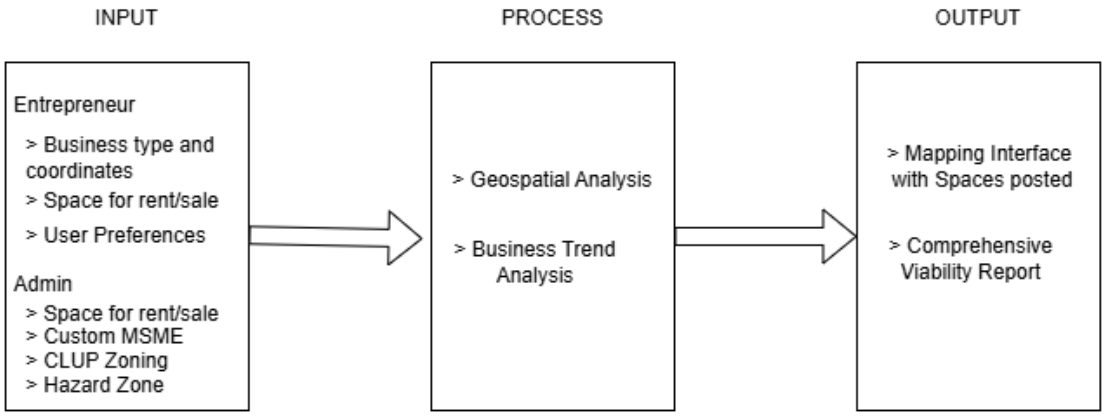


Figure 6. Conceptual Framework

Functional Requirements

1. Uses a web-based mapping interface to visualize locations of the business, infrastructure facilities and at-risk areas.
2. Seeks to assess spatial market demand and competition from building density and road networks.
3. Determines a suitability score for locations based on multiple criteria decision-making process)
4. Samples city planning information and data (such as City Land Use Plan [CLUP]) to verify zoning compliance.
5. Recognizes underserved target markets and business opportunities.
6. Evaluates environmental hazards (including flood hazard area)
7. Generates automated feasibility reports leveraging on the AI powered reporting module.
8. Hybrid Evaluation – supports users to analyze and compare up to 2 types of business.
9. Centralizes information and data of the things of everyday life and information from analysis.
10. Offers a responsive website that works on mobile devices.
11. Takes note of environmental risks (e.g. floodplains) and overlays them over an overall suitability map; these are color coded and automatically modify the suitability score.
12. Analyzes spatial dynamics and markets to proactively determine the types of businesses that can be and are profitable, and to speed up the process of discovering businesses.

Non- Functional Requirements

1. The system must display maps, analysis results, and reports through a web-based dashboard for easy access.
2. The system must be accessible through modern web browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.
3. The system must be compatible with desktop, tablet, and mobile devices.

4. The system interface must be simple, clear, and user-friendly for entrepreneurs.
5. The system must ensure accurate and reliable analysis of spatial and business data.
6. The system must maintain stable performance when processing multiple user requests.
7. The system must ensure data security and protect stored information from unauthorized access.
8. The system must provide fast processing of user queries and analysis results.

Use Case Diagram

The Use Case Diagram of the MarketScope system displays the interrelationships among the users of MarketScope system and the activities that can be performed in the system for analyzing business locations and to generate feasibility reports. The primary user, who is the entrepreneur or MSME owner, uses the system through selecting a location, choosing a business category and viewing the analysis results. The system takes in the user's input and performs geospatial analysis, market demand and competition analysis, compliance with zoning and hazard risk assessment. The feasibility report is then created by the AI module based on the processed data. The system supports users in exploring maps and understanding the suitability scores, and allows for the reports to be shared and offers a comprehensive and user friendly decision support process.

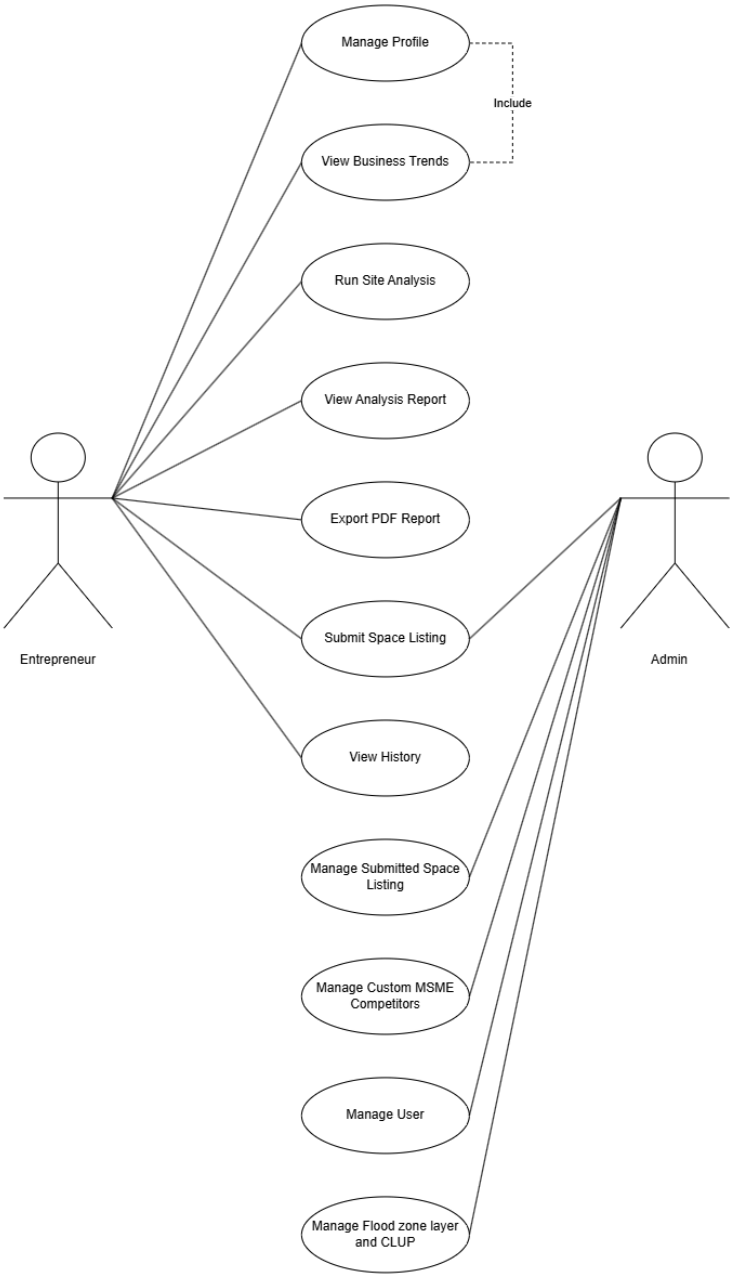


Figure 7. Use Case Diagram of MarketScope

Context Flow Diagram

Figure 8 is a Level 0 Data Flow Diagram (Context Diagram) of MarketScope Geospatial, Business and Business Trend Analysis Engine as the central processing unit where the exchange and interaction with the information and persons is related. The diagram illustrates the functional relationship among the basic analysis engine of the system and the two main external actors involved in the system: Entrepreneur and Admin. The Entrepreneur feeds data into the system – including log-in information, business categories, spatial coordinates – and receives valuable outputs such as viability reports, business trend forecasts, zoning compliance statuses and other information, thus ensuring data-driven decision making. Meanwhile the Admin can choose to control the data deciding the zoning rules within the framework of CLUP, customize space specific MSME parameters and manage space all over while monitoring the spatial analysis, ensuring the spatial analysis is based on authorized data and space. Finally, this architecture is a satisfying mix of spatial query by the user and administrative control, allowing that MSME can confidently be the one who makes informed decisions about where to locate.

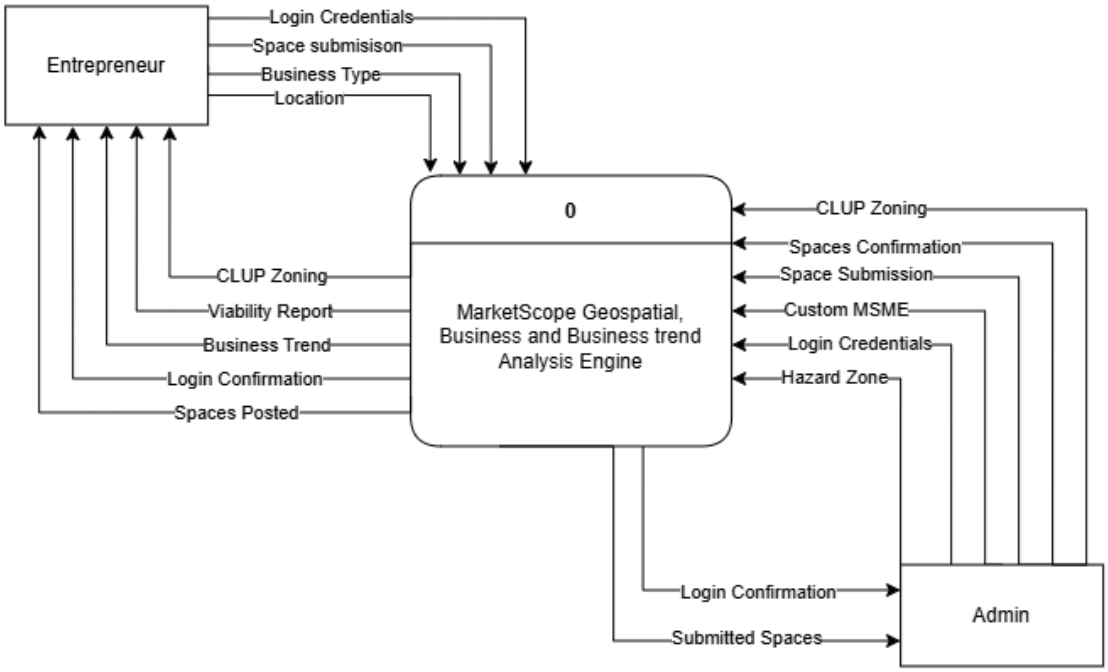


Figure 8. Context Flow Diagram of MarketScope

Data Flow Diagram

The Data Flow Diagram shows the main processes of the MarketScope application and also shows data flow throughout the application across all platforms of the application. The parameters that will be fed into the system will be selected location, business sector and options that will be set by the users for analysis. This information will be passed through multiple system components such as geospatial mapping, market demand and competition analysis, zoning analysis, hazard risk analysis, creation of appropriate scores and report generation through AI. Business locations, infrastructural data, CLUP zoning data, and hazard data will be included in the database that will be accessed by the system. After processing the data by the application, the output will be produced to the user and it will consist of visualization, appropriate scores, market gap analysis, zoning status, hazard risk assessment and feasibility reports download.

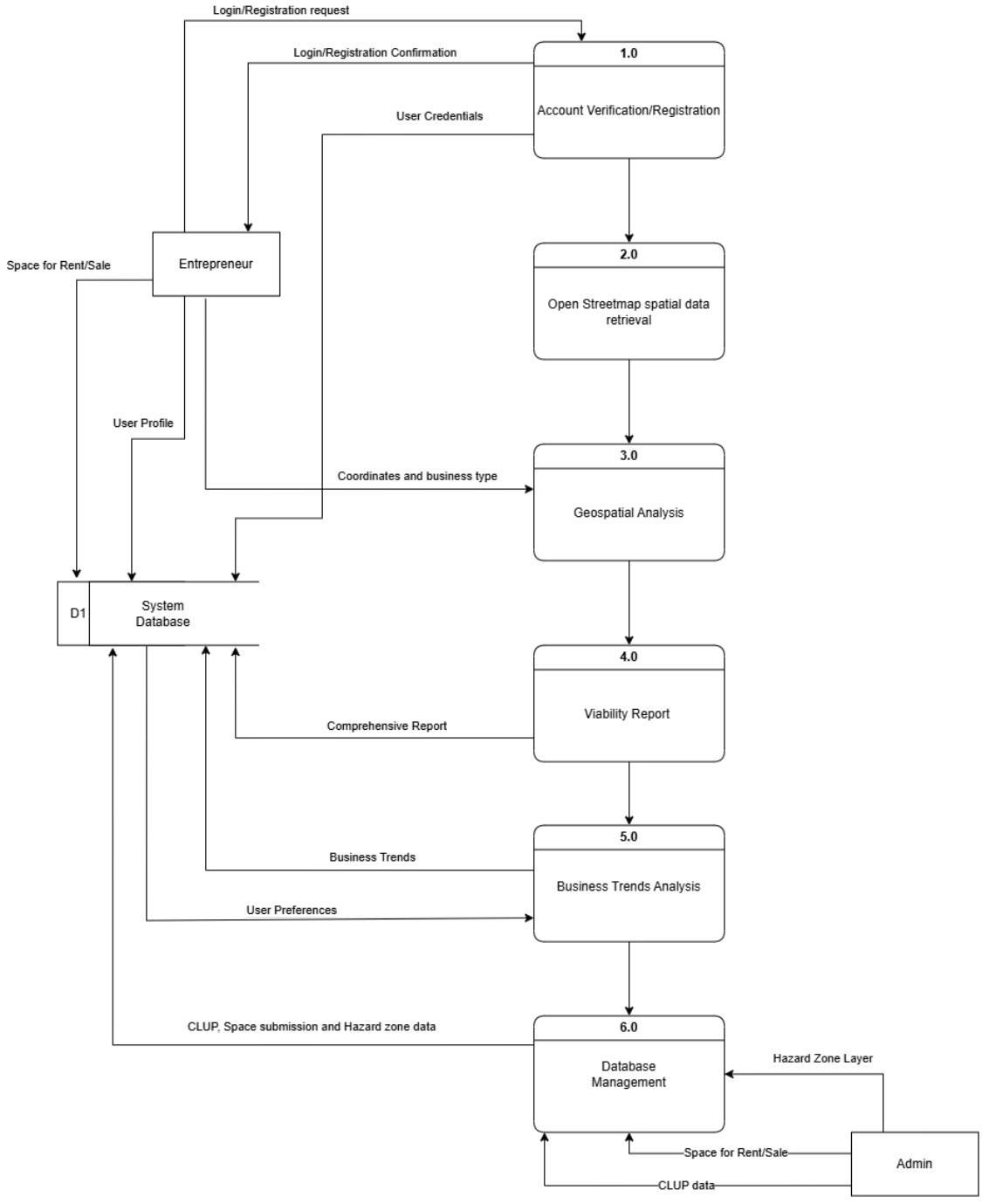


Figure 9. Data Flow Diagram of MarketScope

Entity Relationship Diagram

The Entity Relationship Diagram (ERD) represents the architecture of the MarketScope system and shows the relationship between the important data entities. The database contains important information such as user information, business categories selected by the user, business location, zoning data, hazard risk zones, suitability score, and feasibility report generated. User entity will contain the account information and will be connected to the analysis records created by a user. Each analysis record will contain the business location selected, the business category selected, the zoning assessment, hazard risk assessment, and overall suitability score. Conversely, the category of business will be a business type entity that will show the different MSME categories that can be analysed with the platform. Finally, the spatial data entity will include data such as the existing businesses, infrastructure data, CLUP zoning data, and flood-prone zones.

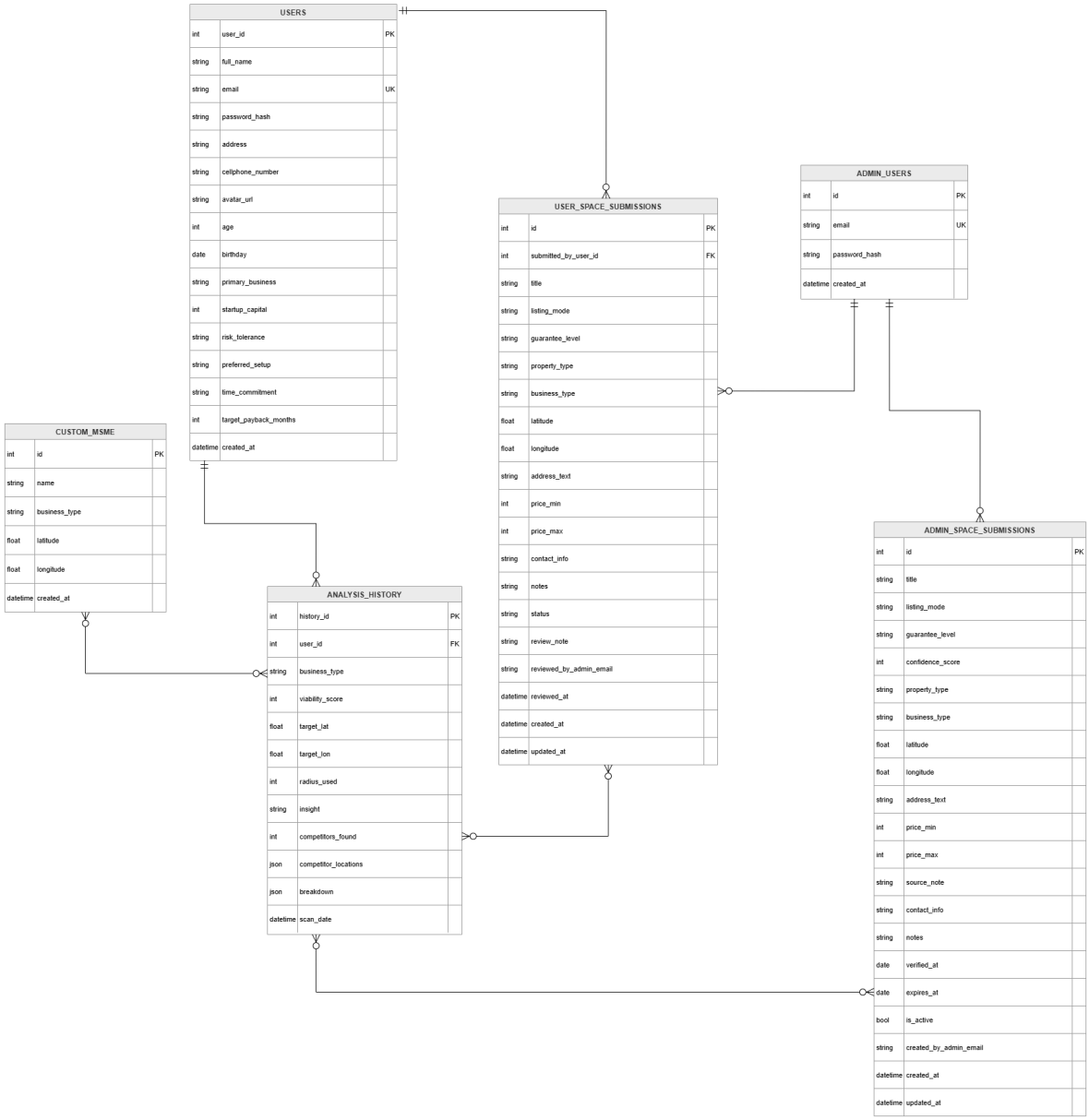


Figure 10. Entity Relationship Diagram of MarketScope

Data Dictionary

The Data Dictionary defines the structure, type, and purpose of all data used in the MarketScope system, ensuring consistency and accurate data management for analysis and reporting.

Table 2. Data Dictionary

USERS

Column name	Datatype	Constraint	Description
user_id	Integer	Primary Key	Unique identifier for the user.
full_name	Varchar(255)	Not Null	The complete name of the entrepreneur.
email	Varchar(255)	Unique, Not Null	Email address used for authentication.
password_hash	Text	Not Null	Encrypted (bcrypt) password string.
address	Text	Not Null	Physical address of the user.
cellphone_number	Varchar(32)	Nullable	Contact number of the user.
primary_business	Varchar(120)	Nullable	The user's main industry of interest.
startup_capital	Integer	Nullable	The estimated budget of the user for business planning.
risk_tolerance	Varchar(20)	Nullable	User's risk preference for system recommendations.
created_at	Timestamp	Default: CURRENT	Date and time the account was created.

Admin User

Column name	Datatype	Constraint	Description
id	Integer	Primary Key	Unique identifier for the admin.
email	Varchar(255)	Unique, Not Null	Admin email address for login.
password_hash	Text	Not Null	Encrypted password string.
created_at	Timestamp	Default: CURRENT	Date and time the admin account was created.

Analysis History

Column name	Datatype	Constraint	Description
history_id	Integer	Primary Key	Unique identifier for the user.
user_id	Varchar(255)	Not Null	The complete name of the entrepreneur.
business_type	Varchar(255)	Unique, Not Null	Email address used for authentication.
viability_score	Text	Not Null	Encrypted (bcrypt) password string.
target_lat	Text	Not Null	Physical address of the user.
target_lon	Varchar(32)	Nullable	Contact number of the user.

radius_used	Varchar(120)	Nullable	The user's main industry of interest.
insight	Integer	Nullable	The estimated budget of the user for business planning.
competitors_found	Varchar(20)	Nullable	User's risk preference for system recommendations.
competitor_locations	Timestamp	Default: CURRENT	Date and time the account was created.
breakdown	JSONB	Nullable	Detailed JSON object storing sub-scores (zoning, hazard).
scan_date	JSONB	Default: CURRENT	When the analysis was executed.

User Space Submission

Column	Datatype	Constraint	Description
name			
id	Integer	Foreign Key	Unique identifier for the listing.
submitted_by_user_id	Integer	Foreign Key	Links the listing to the submitting users(user_id).
title	Varchar(255)	Not Null	The display name of the space listing.

listing_mode	Varchar(12)	Not Null	Identifies if the space is for Rent or Sale.
property_type	Varchar(80)	Nullable	Defines if it is an empty lot, commercial space, etc.
latitude	Double Precision	Not Null	Latitude coordinate of the available space.
longitude	Double Precision	Not Null	Longitude coordinate of the available space.
price_min	Integer	Nullable	Minimum price range in PHP.
status	Varchar(20)	Default: CURRENT	Workflow state (pending, approved, rejected).
reviewed_by_admin_email	Varchar(255)	Nullable	Admin who processed the submission.

Technologies, Concepts, and Theories

The MarketScope system incorporates the concepts of geospatial analytics and artificial intelligence to deliver information-driven decision-making in determining optimal sites for small business. OpenStreetMap data and local government data sets are used for gathering geographic data, including geographic position of the various businesses, roadways, infrastructure, zoning information, and disaster-related risk factors. The data are then processed with a web-based application, which utilizes mapping methods, a central database and a report generator based on artificial intelligence.

The geospatial analysis and Multi-Criteria Decision Analysis (MCDA) have been widely applied in location-oriented decision making support systems, taking into account multiple criteria such as demand, competition, accessibility and environment hazards. MarketScope uses geo-analysis and MCDA to calculate the suitability score on the basis of spatial association and indirect criteria like building density, or proximity to critical infrastructure. By doing so, MarketScope can identify undervalued areas, identify market gaps and provide informed advice on business location.

The system is also based on three theoretical concepts. Integrating spatial data, spatial data processing modules and interaction with users into one decision support system is called General Systems Theory. Data Driven Decision Making theory involves the significance of the usage of data analyzed and reported to make decisions regarding entrepreneurship and proper decision making. The theory of sustainable development helps the above-mentioned theories to balance urban development, by placing businesses in accordance with the zoning laws and policies.

Comparative Evaluation

GIS-AHP (Analytic Hierarchy Process)

GIS-AHP is a structured decision making approach that breaks down spatial problems into a hierarchical matrix of criteria where every criterion is mathematically weighted by the use of pairwise comparisons, and assigns specific numeric weights to each criterion, for example road proximity and competitor density. Its deductive structure matched perfectly with the local government unit policies, which translate deterministic zoning laws into transparent weights that are linear and allow for legally explainable and very accurate spatial modeling [27]. AHP has also been recently proved by geospatial studies to be the best available technique alone for urban planning, since it imposes a system of rigid environmental and regulatory constraints without data distortion [28].

TOPSIS Algorithm

The way TOPSIS works is to calculate the Euclidean distance between geographic alternatives and a theoretical Positive Ideal Solution (PIS) and a Negative Ideal Solution (NIS) and then use a Closeness Coefficient to directly determine the rank of each location by how close it is to the ideal [29]. Although TOPSIS is very computation-efficient, it was added to illustrate an important mathematical weakness of spatial analytics called "rank reversal. As Yang et al. find, this is well known and unresolved in practice: As more alternatives are added or removed, the order of the existing alternatives changes, and the reliability of TOPSIS is significantly degraded [30]. To validate the reason why a purely geometric distance algorithm is unstable for a dynamic city map with new business coordinates constantly being added and evaluated, Benchmarking TOPSIS can be performed.

FUZZY Logic

To evaluate the system's performance on the problems of spatial distance decay and boundary uncertainty, Fuzzy Logic was introduced. Fuzzy Logic, in contrast to crisp mathematical logic, in which the distance of a location is either exactly 100 meters or it is not, provides a membership function as a means of simulating gradual, human-like reasoning about distance and proximity, as Tripathi et al. show directly in their work with Fuzzy AHP which models criteria such as accessibility and proximity as continuous degrees, not hard cutoffs [31]. At the same time, this flexibility is what makes Fuzzy Logic not suitable for zoning compliance in particular. There is no middle ground on a Comprehensive Land Use Plan (CLUP): by zoning a parcel as commercial or it is not commercial, or by marking a membership curve, there is no way to blur the line without misrepresenting the law. Because the law considers a certain criterion as absolute, using fuzzy membership with that criterion would add ambiguity to a decision that shouldn't contain any ambiguity, MarketScope uses crisp AHP for that criterion and uses Fuzzy Logic for other continuous variables such as distance decay in the system.

GIS-AHP, TOPSIS, and Fuzzy Logic are prescriptive Multi-Criteria Decision-Making (MCDM) engines. They deterministically calculate suitability scores from spatial criteria and expert-assigned weights, and produce a rank-ordering of candidate locations rather than estimating an unknown quantity. Because no observed "true" ranking of sites exists, such engines cannot be judged by predictive accuracy alone. In site-scanning studies of this type, where a study area is divided into potential sites that are compared on attributes such as demand proxies, accessibility, proximity to activity generators, and competition, the engines are instead evaluated on four properties: the consistency of the judgments used to derive their weights, the agreement between independently formulated engines, their robustness to uncertainty in parameters, and their correspondence with observed conditions. The differences between engines are then tested for statistical significance.

To satisfy the requirement for real rather than synthetic data, the engines were evaluated on secondary data for Davao City, whose larger business population provides enough cases for statistical testing. A regular 300 m grid was laid over the city, and a grid point was retained as a candidate site only if it had at least 88 buildings within 900 m, the level of the 5% least built-up location where an existing establishment operates; points in industrial, military, cemetery, and similar land uses were then excluded. This yielded 2,914 candidate sites. Existing establishments of the 13 MSME types in this study, 5,803 in total, were taken from OpenStreetMap [32] and Foursquare Open Source Places [33], merging records of the two sources that lie within 40 m of each other, or within 150 m with matching names. Population was taken from Meta's High Resolution Population Density Maps [34].

Each site was described by five criteria measured within a 900 m radius, the same radius used by MarketScope. Road Access is the distance to the nearest main road (a cost criterion); Market Demand is the population within the radius (benefit); Traffic Generators is the number of schools, markets, malls, hospitals, and transport terminals (benefit); Competition Density is the number of open establishments of the same type (cost); and Building Density is the building footprint area divided by the area of the circle (benefit). The criteria weights were

adopted from the expert-derived AHP weights of Syaifuddin, Puad, and Rusdian [35], whose criteria (accessibility, population density, number of consumers, competition, and store presence) were mapped to MarketScope's criteria in the order above, giving weights of 0.35, 0.30, 0.20, 0.10, and 0.05, respectively. All three engines used the same candidate sites, criteria, data preparation, and weights, so that any difference in results is attributable to the engines themselves.

The consistency of the weights is a check of the inputs rather than of the engines. AHP derives its weights from human pairwise comparisons, and if any judgment contradicts the others, all subsequent calculations are invalid. Given a reciprocal pairwise comparison matrix A of n criteria, the weight vector w is its normalized principal eigenvector, following Saaty's Analytic Hierarchy Process [36]:

$$A\,\mathbf{w} = \lambda_{\max}\,\mathbf{w}, \qquad a_{kj} = \frac{1}{a_{jk}}, \qquad \sum_{j=1}^{n} w_j = 1$$

The Consistency Index (CI) and Consistency Ratio (CR) follow from the principal eigenvalue λ_max, where RI is the random index, the mean CI of randomly generated matrices (RI = 1.12 for n = 5):

$$CI = \frac{\lambda_{\max} - n}{n - 1}, \qquad CR = \frac{CI}{RI}$$

Judgments are considered logically coherent when CR < 0.10. The implementation of Eqs. (1) and (2) was verified against the published retail location matrix of Akalin, Turhanbi, and Sahin [37], reproducing their weights (0.526, 0.124, 0.077, and 0.273) and their CR of 0.04. The weights adopted for this benchmark were published as final values without the underlying pairwise matrix, so their CR cannot be recomputed; deriving a matrix from the weights themselves would produce a CR of zero by construction and would carry no information about the consistency of real judgments. Because the CR describes

the weighting step that all three engines share, and does not distinguish between them, it is reported here rather than in the comparative table. It will be reported once the weights are elicited from local experts.

The three engines share one preprocessing step and differ only in how they standardize and aggregate the criteria. In the following, x_ij is the value of criterion j at site i, w_j is the weight of criterion j, and m is the number of candidate sites. To prevent a few extreme sites, such as one adjacent to a large mall, from compressing the scores of all others, each criterion is first capped at its 1st and 99th percentiles among the candidate sites:

$$\tilde{x}_{ij} = \min\left(\max\left(x_{ij},\, Q_{0.01,j}\right),\, Q_{0.99,j}\right)$$

GIS-AHP applies the weighted linear combination of GIS-based multi-criteria analysis [38]. Each criterion is rescaled linearly to the interval from 0 to 1, with the scale reversed for cost criteria:

$$x'_{ij} = \frac{\tilde{x}_{ij} - \min_i \tilde{x}_{ij}}{\max_i \tilde{x}_{ij} - \min_i \tilde{x}_{ij}} \;\text{(benefit)}, \qquad x'_{ij} = \frac{\max_i \tilde{x}_{ij} - \tilde{x}_{ij}}{\max_i \tilde{x}_{ij} - \min_i \tilde{x}_{ij}} \;\text{(cost)} \quad (4)$$

The suitability score is the weighted sum of the rescaled criteria, multiplied by the constraints c_ik, which equal 1 if site i satisfies constraint k (such as a permitted land use) and 0 otherwise:

$$S_i = \left(\sum_{j=1}^{n} w_j\, x'_{ij}\right) \prod_k c_{ik}$$

TOPSIS ranks sites by their relative distance from an ideal and an anti-ideal site [39]. The criteria are vector-normalized and weighted:

The ideal site A^+ takes the best weighted value of each criterion (the maximum for benefit criteria and the minimum for cost criteria), and the anti-ideal site A^- takes the worst. The distances of each site from both, and its relative closeness C_i, which ranges from 0 at the anti-ideal to 1 at the ideal, are:

$$r_{ij} = \frac{\tilde{x}_{ij}}{\sqrt{\sum_{i=1}^{m} \tilde{x}_{ij}^2}}, \qquad v_{ij} = w_j \, r_{ij}$$

$$D_i^+ = \sqrt{\sum_{j=1}^{n} \left(v_{ij} - A_j^+\right)^2}, \qquad D_i^- = \sqrt{\sum_{j=1}^{n} \left(v_{ij} - A_j^-\right)^2}$$

$$C_i = \frac{D_i^-}{D_i^+ + D_i^-}$$

Fuzzy Logic represents suitability as a degree of membership between 0 and 1 rather than a crisp value [40]. Each criterion is mapped through a monotonic sigmoidal membership function between the control points a_j and b_j, set at the 5th and 95th percentiles of the candidate sites, so that suitability increases gradually rather than in linear steps:

$$t_{ij} = \min\left(\max\left(\frac{\tilde{x}_{ij} - a_j}{b_j - a_j}, 0\right), 1\right), \qquad \mu_{ij} = \begin{cases} \sin^2\left(\frac{\pi}{2} t_{ij}\right) & \text{benefit} \\ \sin^2\left(\frac{\pi}{2} \left(1 - t_{ij}\right)\right) & \text{cost} \end{cases} \tag{9}$$

The suitability score is the weighted aggregation of the memberships:

$$S_i = \sum_{j=1}^{n} w_j \, \mu_{ij}$$

Because suitability is a constructed rather than an observed property, inter-method agreement is determined by comparing the rankings each engine produces for the same decision matrix, following the comparative design of fuzzy AHP and fuzzy TOPSIS by Soloukdar and Parpanchi [41]. Spearman's rank correlation [42] measures this agreement over the m = 2,914 candidate sites,

where d_i is the difference between the two ranks of site i, and ρ = 1 denotes identical rankings:

$$\rho = 1 - \frac{6 \sum_{i=1}^{m} d_i^2}{m\left(m^2 - 1\right)}$$

Kendall's tau [43] counts the pairs of sites that two engines place in the same order (concordant, n_c) and in opposite order (discordant, n_d); its tie-corrected form τ_b was used:

$$\tau = \frac{n_c - n_d}{\frac{1}{2} m\left(m - 1\right)}$$

Since a decision maker acts mainly on the best-ranked sites, the overlap between the top 10% of sites (k = 0.10m) identified by two engines a and b was also measured:

$$O_{ab} = \frac{|T_a \cap T_b|}{k}$$

The agreement reported for each engine is its mean ρ against the other two engines. High agreement indicates that the high-suitability zones are determined by the spatial data rather than by the mathematical structure of any single algorithm.

Robustness is assessed using Monte Carlo weight-perturbation analysis. A systematic review finds that weight-modification analysis is the standard procedure for demonstrating that rankings are not artefacts of arbitrarily precise parameters [44]. Following the simulation approach of Butler, Jia, and Dyer [45], 500 alternative weight vectors were drawn from a Dirichlet distribution centred on the adopted weights with concentration κ = 50. Unlike independent ±10% changes to each weight, this keeps every perturbed vector summing to one; at κ = 50 the largest weight (0.35) varies by about ±0.07:

$$\mathbf{w}^{(s)} \sim \text{Dirichlet}\,(\kappa\,\mathbf{w})\,, \qquad s = 1, \ldots, 500$$

Each engine was rerun with every vector and compared with its baseline. Stability ρ is the mean Spearman correlation between the baseline and perturbed scores, and top-10% retention is the mean share of the baseline top-10% sites T(w) that remain in the top 10%:

$$\bar{\rho}_{\text{stab}} = \frac{1}{500}\sum_{s=1}^{500} \rho\left(S(\mathbf{w}),\, S(\mathbf{w}^{(s)})\right), \qquad R_{\text{top}} = \frac{1}{500}\sum_{s=1}^{500} \frac{\left|T(\mathbf{w}) \cap T(\mathbf{w}^{(s)})\right|}{k} \tag{15}$$

A rank reversal test examines structural integrity: the relative ordering of two locations should be independent of unrelated alternatives, as used by Nazim, Mohammad, and Sadiq [46] and formalized by Wang and Triantaphyllou [47]. The property matters because candidate sets grow over time, as spatial databases are extended with new records. In each of 30 repetitions, 20% of the candidate sites were removed at random, each engine was refitted on the remaining sites (including the percentile caps of Eq. (3)), and the order of P = 20,000 randomly drawn pairs of remaining sites was compared before (S) and after (S′) refitting:

$$RR = \frac{100\%}{|P|} \sum_{(i,l)\in P} \mathbf{1}\left[\text{sgn}\,(S_i - S_l) \neq \text{sgn}\,(S'_i - S'_l)\right]$$

External validity is assessed against the locations where enterprises currently operate, which are the observable evidence of enterprise viability; Mercan and Acıbuca used ROC-AUC against field-verified locations for this purpose [48]. For each MSME type, each engine scored the n_1 existing establishments (p) and n_0 background sites (q), drawn at random from the candidate sites at least 150 m from any establishment of that type. An establishment was not counted as its own competitor. The area under the ROC curve was computed through the Mann–Whitney formulation [49]; it equals the

probability that a randomly chosen establishment scores higher than a randomly chosen background site, where 0.5 denotes random discrimination and 1.0 perfect separation:

$$AUC = \frac{1}{n_1 n_0} \sum_{p=1}^{n_1} \sum_{q=1}^{n_0} \left(\mathbf{1}\left[S_p > S_q\right] + \tfrac{1}{2}\,\mathbf{1}\left[S_p = S_q\right] \right)$$

The hit rate is the share of existing establishments that fall within the suitable class, defined as the top 20% of candidate scores, that is, above their 80th percentile $Q_{0.80}$. An engine without discriminating ability would achieve about 20%:

$$HR = \frac{100\%}{n_1} \sum_{p=1}^{n_1} \mathbf{1}\left[S_p \geq Q_{0.80}\left(S_{\text{candidates}}\right)\right]$$

Because establishments concentrate in dense commercial areas, a single criterion can already separate them from background sites. To show whether combining the criteria adds information, each engine's AUC was compared with that of the strongest single criterion for the same MSME type, and a 95% confidence interval for the difference was obtained from 1,000 bootstrap resamples [50]. A lift whose interval lies entirely above zero indicates added information:

$$\Delta AUC = AUC_{\text{engine}} - \max_j AUC_j^{\text{single}}$$

The main validation retains the Competition Density criterion so that every engine is evaluated as it runs in the system. Because the engines penalize saturated areas in order to find underserved ones, and therefore also penalize

the clustering of well-established commercial areas, a second run without this criterion is reported for comparison. Finally, for the four MSME types with at least 10 closed establishments recorded in Foursquare, Eq. (17) was applied with open establishments as p and closed ones as q, to test whether engine scores relate to enterprise survival.

Every metric was computed separately for each of the 13 MSME types, which serve as the blocks of the statistical comparison. The Friedman test [51] determines whether the k = 3 engines differ across the N = 13 types, where R̄_a is the mean rank of engine a:

$$W = \sum_{t=1}^{N} \operatorname{sgn}\left(y_{1t} - y_{2t}\right) \cdot \operatorname{rank}\left(\left|y_{1t} - y_{2t}\right|\right)$$

Where the Friedman test was significant, each pair of engines was compared with the Wilcoxon signed-rank test [52], which ranks the absolute differences between the two engines' values y_1☐ and y_2☐ across types and sums the ranks with their signs:

$$\chi_F^2 = \frac{12N}{k(k+1)} \sum_{a=1}^{k} \bar{R}_a^2 - 3N(k+1)$$

To control the error rate across the three pairwise comparisons (K = 3), the p-values, sorted in ascending order, were adjusted with Holm's method [53]; a difference was considered significant when the adjusted p < 0.05:

$$\tilde{p}_{(i)} = \max_{l \le i} \; \min\left(1,\; (K - l + 1)\, p_{(l)}\right)$$

All computations were carried out in Python on Google Colaboratory with a fixed random seed of 42, so that the results can be reproduced.

Table 3. Models Comparative Table

Algorithm	Agreement ρ	Stability ρ	Top-10% retention (%)	Rank reversal (%)	ROC-AUC vs real establishments	ΔAUC vs best single criterion	Hit rate (%)
GIS-AHP	0.9931	0.9912	93.9	0.35	0.910	−0.008	84.4
TOPSIS	0.9928	0.9897	92.7	0.23	0.913	−0.005	85.7
Fuzzy Logic	0.9924	0.9917	94.1	0.23	0.903	−0.014	82.4

The performance of the three engines with respect to the evaluation metrics is compared in Table 3. Inter-method agreement was very high: the mean Spearman correlation of each engine with the other two was at least 0.992, and 88% to 98% of the top-10% sites were shared between any two engines, indicating that the high-suitability zones are driven by the spatial data rather than by the structure of any one algorithm. Under 500 Monte Carlo weight perturbations, the mean rank correlation with the baseline exceeded 0.989 for all three engines, demonstrating that the site rankings are not sensitive to fine adjustments of the weights. The top-10% sites were most stable under Fuzzy Logic (94.1%) and GIS-AHP (93.9%), which were statistically indistinguishable, and significantly less stable under TOPSIS (92.7%; adjusted $p < 0.001$ and $p = 0.009$), whose ideal and anti-ideal reference points shift with every change in the weights (Eq. (7)).

The rank reversal test revealed a different structure. When 20% of the candidate sites were removed, the order of the remaining site pairs reversed in 0.35% of cases under GIS-AHP, compared with 0.23% under both TOPSIS and Fuzzy Logic; the difference is statistically significant (adjusted $p \leq 0.001$). The min-max rescaling of GIS-AHP (Eq. (4)) depends on the most extreme sites in the candidate set, so removing those sites changes every score, whereas the vector norms of TOPSIS (Eq. (6)) and the percentile control points of Fuzzy Logic (Eq. (9)) are aggregate properties of the set and change less. In practical terms,

about one site pair in 300 changes order, which is too few to alter a recommendation.

All three engines discriminated well between existing establishments and randomly distributed background sites, with mean ROC-AUC values of 0.913 for TOPSIS, 0.910 for GIS-AHP, and 0.903 for Fuzzy Logic, all far above the 0.5 threshold of random discrimination. The Friedman test confirmed that the engines differ ($\chi^2 = 10.31$, $p = 0.006$): GIS-AHP and TOPSIS were statistically tied (adjusted $p = 0.89$), and both outperformed Fuzzy Logic (adjusted $p = 0.002$ and $p = 0.012$). The hit rates follow the same order: 85.7% of existing establishments fell within the suitable class under TOPSIS, 84.4% under GIS-AHP, and 82.4% under Fuzzy Logic, against about 20% expected by chance. However, none of the engines outperformed the strongest single criterion on average (ΔAUC between -0.005 and -0.014), and only for bakeries did the combined score add significant information. Within the study area, the criteria largely measure the same underlying property, proximity to commercial centres, so combining them adds little discriminating power beyond the strongest one.

The competition criterion behaved contrary to its assumed direction. Scored on its own as a cost, competition density yielded ROC-AUC values between 0.04 and 0.24, far below random discrimination, indicating that established enterprises cluster near competitors of the same type. Removing this criterion raised the mean ROC-AUC only slightly (GIS-AHP 0.921, TOPSIS 0.918, Fuzzy Logic 0.914) because of its low weight (0.10). This is consistent with the design intent of the engines, which steer new entrants away from saturated areas rather than imitate existing clustering. No engine distinguished establishments that remained open from those that closed (mean AUC 0.43 to 0.45), so the results support no claim regarding enterprise survival.

Taken together, the findings indicate that all three engines satisfy the requirements of agreement and robustness and discriminate far better than chance against real establishments. GIS-AHP is the only engine among the best performers in validation accuracy, hit rate, and weight stability simultaneously; its slightly higher rank-reversal rate has no practical consequence; and its weighted

linear structure (Eq. (5)) allows the contribution of each criterion to a site's score to be shown directly to MSME owners. GIS-AHP is therefore retained as the suitability engine of MarketScope. These results, based on 5,803 establishments across 13 MSME types, supersede the earlier small-sample validation. They remain subject to the coverage of OpenStreetMap and Foursquare, which do not record every small business, and to weights adopted from a convenience-store study in another country, which should be replaced with weights elicited from local experts.

Gemini

Gemini is a family of multimodal large language models developed by Google. It is used through the Gemini web application and follows written instructions to produce, summarize, and rewrite text.

ChatGPT

ChatGPT is OpenAI's conversational assistant, built on its GPT family of language models. It is used through the ChatGPT web application and is widely used for writing, summarizing, and explaining text.

Claude

Claude is a family of large language models developed by Anthropic. It is used through the Claude web application and is built to follow detailed written instructions and produce structured text.

The test set has thirteen cases, one for each MSME type supported by MarketScope. Each case is a scan result written in the system's output format, using its AHP weights and scoring rules, and the set covers four high-viability, five moderate-viability, and four low-viability sites with different zoning outcomes and flood hazard classes. The cases were constructed so that conditions missing from the available scans could still be tested. Because every report is judged against its known input, the cases test the accuracy and readability of the generated explanations. They do not validate MarketScope's site scores. For each case, the unchanged MarketScope report prompt was filled with the case's

JSON and pasted into a new chat with memory and personalization turned off. The first reply was saved exactly as given. This produced 39 reports (13 per model). Because the web applications may add their own system instructions and settings, the outputs may differ from those of the corresponding APIs. This is a limitation of the study.

The evaluation combines automatic and human measures, in line with prior work on simplification that pairs readability indices with human judgment [54], [55]. Readability was measured with the Flesch-Kincaid Grade Level (FKGL) [56] and the Flesch Reading Ease (FRE) score [57]. A lower FKGL and a higher FRE mean easier text. Because FKGL can be improved sharply with little change to other metrics [58], it is interpreted together with the other indicators and not used alone. SARI [59] was not used, since it requires reference simplifications and this task has none. Compliance with the prompt was checked automatically for each report: whether the eight required headings appear in order, whether the score is written as "Viability score: N/100", whether JSON or code fences are avoided, whether every number in the report appears in the input, whether zoning and flood warnings are retained, and report length. Human raters who do not know which model wrote each report [N raters; update after data collection] score each one on a 5-point Likert agreement scale (1 = strongly disagree, 5 = strongly agree) for meaning preservation, simplicity, and fluency. This is the format used in [60] and [61]; the latter follows the criteria of [62]. Two researcher-designed items (clarity of warnings and appropriateness of the recommendation and next actions) and a count of errors are added. Scores are summarized as mean and standard deviation per model, compared with a Friedman test [63] with Wilcoxon signed-rank follow-up tests [64], and rater agreement is reported using weighted Cohen's kappa [65].

For each report, the number of words (W), sentences (S), and syllables (Y) was counted with the Python textstat library (v0.7.13). Before counting, markdown symbols were removed and each line of a list or table was ended with a period, so that each such line counts as one sentence. The average sentence

length (ASL, words per sentence) and the average number of syllables per word (ASW) are:

$$\mathrm{ASL} = \frac{W}{S}, \qquad \mathrm{ASW} = \frac{Y}{W}$$

FKGL [56] and FRE [57] are then computed from them as:

$$\mathrm{FKGL} = 0.39\,\mathrm{ASL} + 11.8\,\mathrm{ASW} - 15.59$$

$$\mathrm{FRE} = 206.835 - 1.015\,\mathrm{ASL} - 84.6\,\mathrm{ASW}$$

The values in Table I are the mean and the sample standard deviation of these per-report values over the n = 13 reports of each model:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

As a worked example, the Claude report for case S06 has W = 541, S = 65, and Y = 819. Then ASL = 541 / 65 = 8.323, ASW = 819 / 541 = 1.514, FKGL = 0.39 × 8.323 + 11.8 × 1.514 − 15.59 = 5.52, and FRE = 206.835 − 1.015 × 8.323 − 84.6 × 1.514 = 70.31.

Each compliance check scores a report 1 if it passes and 0 if it fails, and Table II gives the sum over each model's 13 reports. The number check compares every number in a report, apart from heading numbers, with the numbers in the case's input JSON and with the scale values 0 to 8, 25, and 100, and any number that is not found is verified by hand. The warning checks apply only where they are

relevant: the flood check to the six cases with a flood hazard score of 6, 12, or 18, and the zoning check to the four cases with a zoning score of 0. The per-report counts, every formula, and the script that produced them are provided in Appendix [A: MarketScope_Evaluation_Computation.xlsx and MarketScope_metrics_script.py].

For the human ratings, the Friedman statistic [63] over k = 3 models and n rated cases, with R_j the rank sum of model j, is:

$$\chi_F^2 = \frac{12}{nk(k+1)} \sum_{j=1}^{k} R_j^2 - 3n(k+1)$$

and agreement between two raters is measured with weighted Cohen's kappa [65], where w_ij are the disagreement weights, O_ij the observed counts, and E_ij the counts expected by chance:

$$\kappa_w = 1 - \frac{\sum_{i,j} w_{ij} O_{ij}}{\sum_{i,j} w_{ij} E_{ij}}$$

Table 4. Large Language Model Comparative Table

Model	Words (W)	ASL (words per sentence)	ASW (syllables per word)	FKGL (lower = easier)	FRE (higher = easier)
Gemini	419.2 (71.4)	11.1 (2.1)	1.70 (0.06)	8.9 (1.2)	51.3 (6.2)
ChatGPT	746.4 (47.3)	9.2 (0.5)	1.81 (0.05)	9.4 (0.6)	44.1 (4.3)

Claude	509.8 (42.3)	8.3 (0.7)	1.53 (0.02)	5.7 (0.4)	68.7 (1.9)

Table 5. Large Language Model Comparative Table

Model	8 numbered headings in order	"Viability score: N/100" format	No JSON or code fences	Reports with numbers not in the input	Flood warning kept (6 hazard cases)	Zoning warning kept (4 not-allowed or unmapped cases)
Gemini	13	9	13	0	6	4
ChatGPT	13	13	13	0	6	4
Claude	12	13	13	0	6	4

Table 6. Prompt Result Rating

The automatic results show clear differences in readability and in following the prompt, but not in factual accuracy. Claude's reports had the lowest FKGL (5.7) and the highest FRE (68.7), and its FKGL was the lowest in all 13 cases, which is close to a sixth-grade reading level. Gemini (8.9) and ChatGPT (9.4) were closer to a ninth-grade level, and their FRE scores of 51.3 and 44.1 fall in the fairly difficult to difficult range of the Flesch scale. This matters because the prompt asks for simple wording for people new to business. The formulas show where the gap comes from. Claude had both the shortest sentences (8.3 words) and the shortest words (1.53 syllables per word). ChatGPT's sentences were shorter than Gemini's (9.2 against 11.1 words), but its words were the longest (1.81 syllables per word), which gave it the highest FKGL. ChatGPT's reports

were also the longest (about 750 words), despite the instruction to keep them concise. Gemini's reports were the shortest but varied the most in length.

All three models reproduced the input numbers without inventing any, and all kept the zoning and flood warnings, including the case where the site lies outside every mapped zone. These checks therefore did not separate the models. The differences were in instruction following: Gemini did not use the required "Viability score: N/100" wording in four of 13 reports, and one Claude report left out the heading numbers. These results are descriptive only. They come from 13 constructed test cases, from web chat interfaces, and from surface-level metrics, which cannot judge whether an explanation is correct and helpful [58].

System Testing and Implementation

System testing and implementation stage of the MarketScope system ensures the proper functioning of the application and its readiness for use. Testing includes the following steps: verifying the efficiency of the application in conducting all its tasks, geospatial analysis of market demand and competition, zone checking, hazard evaluation, calculation of suitability scores and production of feasibility reports. Additionally, testing will be used to verify that map visualization is accurate and the output from AI reporting is useful and accurate.

Implementation includes a series of steps, such as establishing the web platform and its functionality, adding geospatial layers to the system like business locations, zones, hazards, and other data, and making the system accessible to users via web browsers. Users are trained to use the system properly and data is kept and secured properly.

The MarketScope system is designed for success and tested and implemented during the system testing phase. Testing includes checking that the application can carry out all of its activities efficiently, such as geospatial analysis of market demand and competition, zone checking, hazard evaluation and calculation of suitability scores as well as the generation of feasibility reports.

Additionally, testing will confirm the accuracy of the map visualization and the usefulness and accuracy of the output of AI reporting.

Implementation means having a well-organized process, begin with setting up the web platform and its function, then integrate geospatial layers to the system such as business location, zones, hazards and other information, till make the system available to the users through web browsers. Users are trained to use the system properly and data is securely stored and managed.

System Maintenance

System maintenance will be carried out to guarantee the long term reliability, accuracy and usability of MarketScope. This will involve keeping up to date business location records, infrastructure data, CLUP zoning information and hazard risk maps as new or revised datasets become available. This will also include the correction of errors, performance enhancements, improving security and making improvements to the user interface, based on feedback. As the system is based on the spatial and local government data and new datasets will be released over time, these will need to be updated to ensure analysis results remain accurate. Ongoing maintenance will help ensure that Market Scope is useful, relevant, and effective as a decision support tool for MSMEs and entrepreneurs in Panabo City.

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